

Hands-on Lab: CREATE, ALTER, TRUNCATE, DROP

Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data using the phpMyAdmin graphical user interface (GUI) tool in the MySQL database service.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab, you will use MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Objectives

After completing this lab, you will be able to use phpMyAdmin with MySQL to:

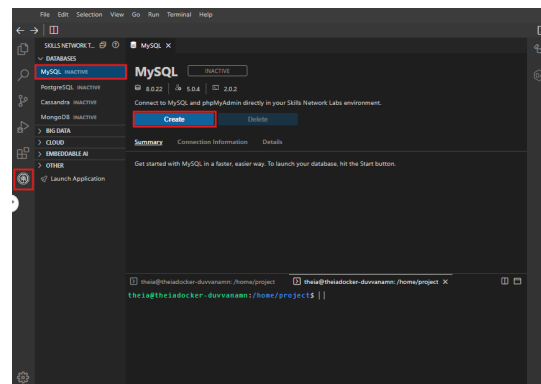
- Create a database.
- Create a new table in a database.
- Add, delete, or modify columns in an existing table.
- Remove all rows from an existing table without deleting the table itself.
- Delete an existing table in a database.

Task 1: Create a database

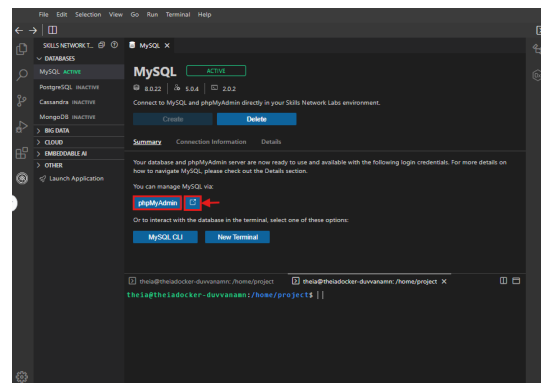
Follow the steps below to create a new database in the phpMyAdmin GUI of MySQL.

1. Click on **Skills Network Toolbox**. In the **Database** section, click **MySQL**.

To start the MySQL, click **Create**.



2. Once **MySQL** has started, click the **phpMyAdmin** button to open **phpMyAdmin** in the same window. Alternatively, click the **toggle button** next to the **phpMyAdmin** button to open phpMyAdmin in a new browser tab.



3. You will see the phpMyAdmin GUI tool.

The screenshot shows the phpMyAdmin interface for a MySQL server (mysql:3306). The left sidebar contains a tree view with 'New', 'information_schema', 'mysql', 'performance_schema', 'sakila', and 'sys'. The main content area is divided into three panels: 'General settings', 'Appearance settings', and 'Database server'. The 'General settings' panel shows 'Server connection collation' set to 'utf8mb4_unicode_ci'. The 'Appearance settings' panel shows 'Language' set to 'English' and 'Theme' set to 'pmahomme'. The 'Database server' panel lists server details: 'Server: mysql via TCP/IP', 'Server type: MySQL', 'Server connection: SSL is not being used', 'Server version: 8.0.22 - MySQL Community Server - GPL', 'Protocol version: 10', 'User: root@172.18.0.2', and 'Server charset: UTF-8 Unicode (utf8mb4)'. The 'Web server' panel lists: 'Apache/2.4.38 (Debian)', 'Database client version: libmysql - mysqlnd 7.4.15', 'PHP extension: mysqli', 'curl', 'mbstring', and 'PHP version: 7.4.15'. The 'phpMyAdmin' panel lists version information and links to documentation, official homepage, contribute, get support, list of changes, and license.

4. In the tree view, click **New** to create a new empty database. Then, enter `mysql_learners` as the name of the database, leave the default `utf8` encoding, and click **Create**.
UTF-8 is the most commonly used character encoding for content or data.

The screenshot shows the 'Databases' tab in phpMyAdmin. The 'Create database' form is visible, with 'mysql_learners' entered in the 'Database' field and 'utf8_general_ci' selected in the 'Collation' dropdown. The 'Create' button is visible. Below the form is a table listing existing databases:

Database	Collation	Master replication	Action
information_schema	utf8_general_ci	✓ Replicated	ⓘ Check privileges
mysql	utf8mb4_0900_ai_ci	✓ Replicated	ⓘ Check privileges
performance_schema	utf8mb4_0900_ai_ci	✓ Replicated	ⓘ Check privileges
sys	utf8mb4_0900_ai_ci	✓ Replicated	ⓘ Check privileges

Total: 4

⬅️ ☐ Check all With selected: Drop

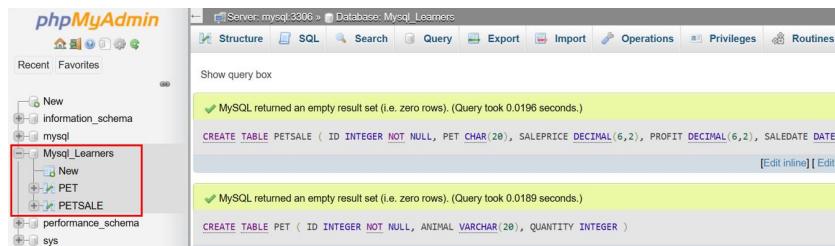
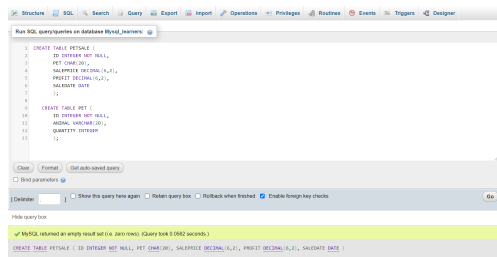
Task 2a : CREATE statement

Now, you will use the `CREATE` statement to create two new tables.
Follow the instructions to complete this task.

1. You need to create two tables, `PETSALE` and `PET`. To create the two tables, copy the code below and paste it into the text area of the `SQL` tab. Click **Go**.

```
CREATE TABLE PETSALE (  
  ID INTEGER NOT NULL,  
  PET CHAR(20),  
  SALEPRICE DECIMAL(6,2),  
  PROFIT DECIMAL(6,2),  
  SALEDATE DATE  
);
```

```
CREATE TABLE PET (
  ID INTEGER NOT NULL,
  ANIMAL VARCHAR(20),
  QUANTITY INTEGER
);
```



Task 2b: INSERT statement

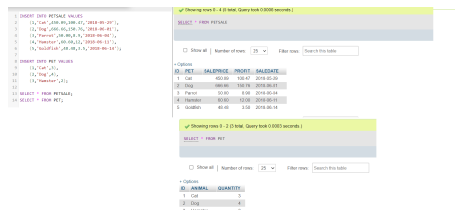
Now, insert some records into the two newly created tables. You can also add SELECT statements to print the contents of the tables once they are loaded with data.

Copy the code below and paste it into the text area of the sql tab. Then, click Go.

```
INSERT INTO PETSale VALUES
(1, 'Cat', 458.89, 188.47, '2018-05-29'),
(2, 'Dog', 666.66, 158.76, '2018-06-01'),
(3, 'Parrot', 58.88, 8.9, '2018-06-04'),
(4, 'Hamster', 68.68, 12, '2018-06-11'),
(5, 'Goldfish', 48.48, 3.5, '2018-06-14');
```

```
INSERT INTO PET VALUES
(1, 'Cat', 3),
(2, 'Dog', 4),
(3, 'Hamster', 2);

SELECT * FROM PETSale;
SELECT * FROM PET;
```



Task 3: ALTER statement

In this exercise, you will use the ALTER statement to add, delete, or modify columns in the existing tables.

1. Adding a column

Add a new column named QUANTITY to the PETSale table and display the altered table. For this, copy the code below and paste it into the text area of the SQL page. Click Go.

```
ALTER TABLE PETSale
ADD COLUMN QUANTITY INTEGER;

SELECT * FROM PETSale;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

ALTER TABLE PETSale ADD COLUMN QUANTITY INT(80);

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 4 (5 total). Query took 0.0005 seconds.)

SELECT * FROM PETSale;

Show all | Number of rows: 25 | Filter rows: Search this table

Options

ID	PET	SALEPRICE	PROFIT	SALEDATE	QUANTITY
1	Cat	450.00	100.00	2018-05-20	9
2	Dog	666.66	100.00	2018-06-01	3
3	Parrot	50.00	0.00	2018-06-04	2
4	Hamster	60.60	12.00	2018-06-11	6
5	Goldfish	48.48	2.00	2018-06-14	24

Now update the newly added QUANTITY column of the PETSale table with some values and show all the table records. Copy the code below and paste it into text area of the SQL page. Click Go.

```
UPDATE PETSale SET QUANTITY = 9 WHERE ID = 1;
UPDATE PETSale SET QUANTITY = 3 WHERE ID = 2;
UPDATE PETSale SET QUANTITY = 2 WHERE ID = 3;
UPDATE PETSale SET QUANTITY = 6 WHERE ID = 4;
UPDATE PETSale SET QUANTITY = 24 WHERE ID = 5;
SELECT * FROM PETSale;
```

UPDATE PETSale SET QUANTITY = 9 WHERE ID = 1;
UPDATE PETSale SET QUANTITY = 3 WHERE ID = 2;
UPDATE PETSale SET QUANTITY = 2 WHERE ID = 3;
UPDATE PETSale SET QUANTITY = 6 WHERE ID = 4;
UPDATE PETSale SET QUANTITY = 24 WHERE ID = 5;
SELECT * FROM PETSale;

ID	PET	SALEPRICE	PROFIT	SALEDATE	QUANTITY
1	Cat	450.00	100.00	2018-05-20	9
2	Dog	666.66	100.00	2018-06-01	3
3	Parrot	50.00	0.00	2018-06-04	2
4	Hamster	60.60	12.00	2018-06-11	6
5	Goldfish	48.48	2.00	2018-06-14	24

2. Deleting a column

Delete the PROFIT column from the PETSale table and show the altered table. Copy the code below and paste it into the text area of the SQL page. Click Go.

```
ALTER TABLE PETSale
DROP COLUMN PROFIT;
SELECT * FROM PETSale;
```

Browse | Structure | SQL | Search | Insert

Run SQL query/queries on table Mysql_learners.PETSale:

1 ALTER TABLE PETSale
2 DROP COLUMN PROFIT;
3
4 SELECT * FROM PETSale;

Options

ID	PET	SALEPRICE	SALEDATE	QUANTITY
1	Cat	450.00	2018-05-20	9
2	Dog	666.66	2018-06-01	3
3	Parrot	50.00	2018-06-04	2
4	Hamster	60.60	2018-06-11	6
5	Goldfish	48.48	2018-06-14	24

Show all | Number of rows: 25 | Filter rows: Search this table

3. Modify a column

Change the data type to VARCHAR(20) type of the column PET of the table PETSale and show the altered table. Copy the code below and paste it into the text area of the SQL page. Click Go.

```
ALTER TABLE PETSale
MODIFY PET VARCHAR(20);
SELECT * FROM PETSale;
```

You can click on the table name PETSale in the tree structure on the left and then click on the Structure tab in the interface. You can then see the table structure shows the modified column data type, as shown in the image below.

The screenshot shows a database management interface with a menu bar (Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations) and a toolbar. The main area is titled 'Run SQL query/queries on table Mysql_learners.PETSALE:'. Below this, there is a text area for SQL queries with the following content:

```
1
2 ALTER TABLE PETSALE CHANGE `PET` `PET` VARCHAR(20);
3
4 SELECT * FROM PETSALE;
```

To the right of the SQL editor, there is a table view showing the structure of the 'PETSALE' table. The table has 5 columns: ID, PET, SALEPRICE, SALEDATE, and QUANTITY. The data is as follows:

ID	PET	SALEPRICE	SALEDATE	QUANTITY
1	Cat	450.09	2018-05-29	9
2	Dog	666.66	2018-06-01	3
3	Parrot	50.00	2018-06-04	2
4	Hamster	60.60	2018-06-11	6
5	Goldfish	48.48	2018-06-14	24

Below the table view, there is a 'Table structure' tab and a 'Relation view' tab. The 'Table structure' tab is active, showing the following table structure:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	ID	int		No	None				Change Drop More
2	PET	varchar(20)	utf8mb4_0900_ai_ci	Yes	NULL				Change Drop More
3	SALEPRICE	decimal(8,2)		Yes	NULL				Change Drop More
4	SALEDATE	date		Yes	NULL				Change Drop More
5	QUANTITY	int		Yes	NULL				Change Drop More

4. Rename a Column

Rename the column `PET` to `ANIMAL` of the `PETSALE` table and show the altered table. Copy the code below and paste it into the text area of the `SQL` page. Click `Go`.

```
ALTER TABLE `PETSALE` CHANGE `PET` `ANIMAL` varchar(20);
SELECT * FROM PETSALE;
```

The screenshot shows the same database management interface as before, but with the SQL query area containing the following code:

```
1 ALTER TABLE `PETSALE` CHANGE `PET` `ANIMAL` varchar(20);
```

Below the SQL editor, there is a green bar indicating the query was successful: 'Showing rows 0 - 4 (5 total, Query took 0.0006 seconds.)'. Below this, there is a table view showing the structure of the 'PETSALE' table. The table has 5 columns: ID, ANIMAL, SALEPRICE, SALEDATE, and QUANTITY. The data is as follows:

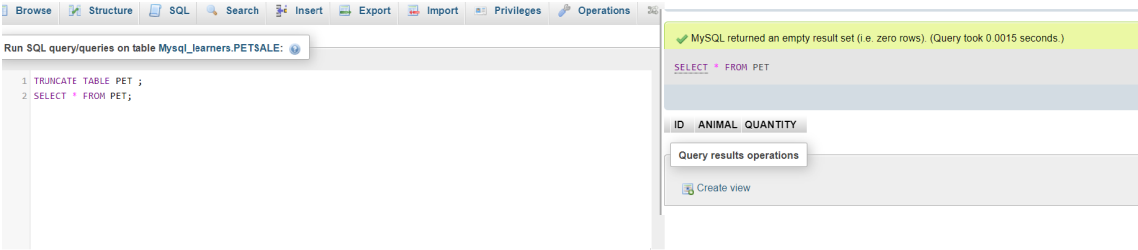
ID	ANIMAL	SALEPRICE	SALEDATE	QUANTITY
1	Cat	450.09	2018-05-29	9
2	Dog	666.66	2018-06-01	3
3	Parrot	50.00	2018-06-04	2
4	Hamster	60.60	2018-06-11	6
5	Goldfish	48.48	2018-06-14	24

Task 4: TRUNCATE statement

In this exercise, you will use the `TRUNCATE` statement to remove all rows from an existing table without deleting it.

Let's remove all rows from the `PET` table and show the empty table. Copy the code below and paste it into the text area of the `SQL` page. Click `Go`.

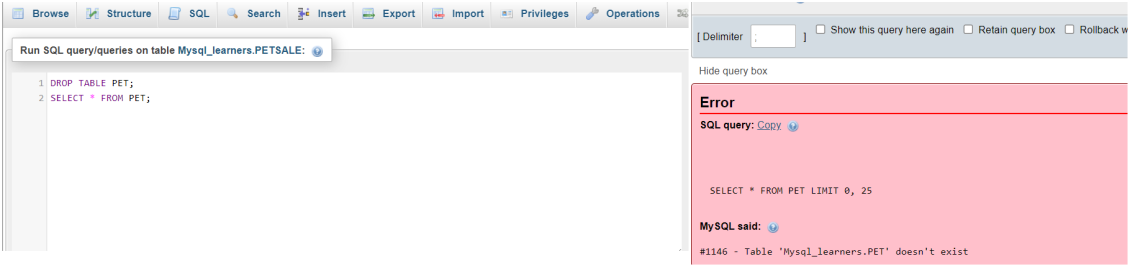
```
TRUNCATE TABLE PET ;
SELECT * FROM PET;
```



Task 5: DROP statement

Finally, you will use the DROP statement to delete an existing table. Let's delete the PET table and verify if the table still exists or not (the SELECT statement should give an error if a table doesn't exist). Copy the code below and paste it into the text area of the sql page. Click Go.

```
DROP TABLE PET;
SELECT * FROM PET;
```



Practice problems

Try the following problems for an enhanced practice of the concepts learned in this lab.

1. Create a new table in the database named TOYS with attributes as ID (integer), Variety (variable length string), and Quantity (integer). Make sure the ID is not Null.
▶ [Click here for the solution](#)
2. Add the below-mentioned entries to the table using the INSERT statement.

ID	Variety	Quantity
1	Chew toy	20
2	Balls	50
3	Bowls	30
4	Foldable bed	40

3. ALTER the length of 'Variety' in the table to 30 characters.
▶ [Click here for the solution](#)
4. TRUNCATE the table 'Toys'
▶ [Click here for the solution](#)
5. DROP the table 'Toys'
▶ [Click here for the solution](#)

Conclusion

Congratulations on successfully completing this lab.

By now, you have learned how to:

- Create a database in phpMyAdmin GUI on MySQL.
- Use the CREATE statement to create new tables in the database.
- Use the INSERT statement to add records to the tables.
- Use the ALTER statement to add, delete, rename, or modify the columns of an existing table.
- Use the TRUNCATE statement to delete the contents of an existing table (but not the table).
- Use the DROP statement to delete an entire table.

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